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The "MERCHANT GUARD" Challenge

Presented by Plub And Kuga

Introduction & Context

“Image-Category Mismatch”

Context

- Currently, merchants **upload images** and **select category** during registration.
- To ensure trust and safety, we must **verify** that a merchant is **who they say they are**.

A photo uploaded by shop owner

The category selected by the shop owner “Services”

Problem Solution Insight Impact

What is the **problem**?

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Problem

Merchants select **wrong category** or upload **irrelevant images**

Impact

- **Operational Bottleneck**
- **High Risk of Fraud & Gaming**
- **Damaged Brand Trust**

Our goal

To ensure that a **merchant is who they say they are.**

Create an **automated solution** to **verify** if the uploaded image match with selected category.

Problem Solution Insight Impact

How to **automated ensure the merchant?**

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Our Solution

Implement an automated **Image-Text Consistency model** to automatically cross-check visual evidence against declared business categories in real-time.

Our Solution

Clean Data: Image-Text Consistency Checking

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80 merchants ▶ **76** merchants

4 merchants
can't link to images

remove 4 merchants

1 merchant **selected**
wrong payment method

Update their method

add label
for evaluation

1 for **Consistent**

0 for **Inconsistent**

Problem | **Solution** | Insight | Impact

Our Solution

Exploratory Data Analysis: Image-Text Consistency Checking

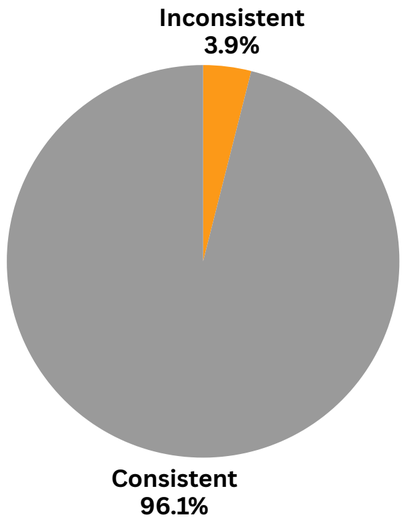
DATA OVERVIEW

76
Merchants

386
Images

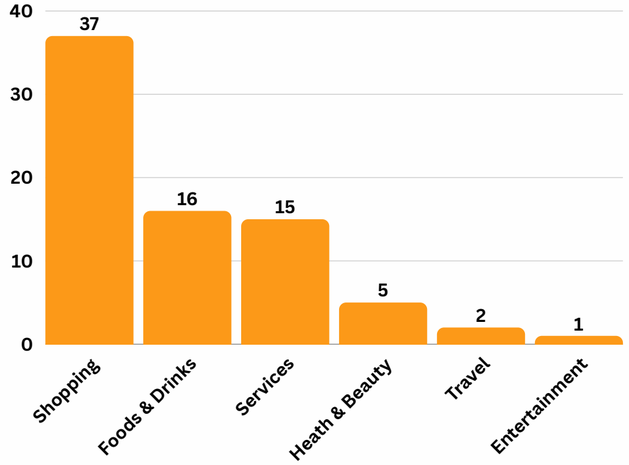
KEY FINDING

Image-Text Consistency Ratio



CONTEXTUAL

Merchant Categories



The Dataset is “Imbalance”

Problem | **Solution** | Insight | Impact

Our Solution

Modeling: Image-Text Consistency Checking

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Current SOTA

CLIP model

- **Visual** Pattern Matching
- **Zero-shot** Classification
- **Recognizes** a "storefront layout"

LLMs model

- Semantic **Reasoning**
- **Logical** Inference
- **Understands** contextual clues

"To See" model

"To Read and Reason" model

Problem **Solution** Insight Impact

Our Solution

Modeling: Hybrid: mCLIP + OCR model

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Hybrid: mCLIP + OCR model

What is mCLIP + OCR?

- **CLIP** - compares the **image to keywords**
- **OCR** - **extract text** from the image
- **Multiligual** - **understand Thai** text descriptions of images

Why mCLIP + OCR?

- Processing happens in near **real-time**
- **Auditable,** can fully control and scale
- **Cost-effective,** doesn't require massive hardware.

How mCLIP + OCR work?

- **Encode** text and image components
- **Predict** Category and Sub-category from image
- **Compare** between selected and predicted category

Problem **Solution** Insight Impact

Our Solution

Modeling: Hybrid: mCLIP + OCR model

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Model Evaluation

Confusion Matrix

Actual \ Predicted	Consistent	Inconsistent
Consistent	47	26
Inconsistent	0	3

Accuracy: 0.6579

Weighted F2-score: 0.6803

Why F2-score?

Cost of Low Recall

- Inconsistent data may exist

Cost of Low Precision

- Need Manual Review

Our Solution

Modeling: Hybrid: mCLIP + OCR model

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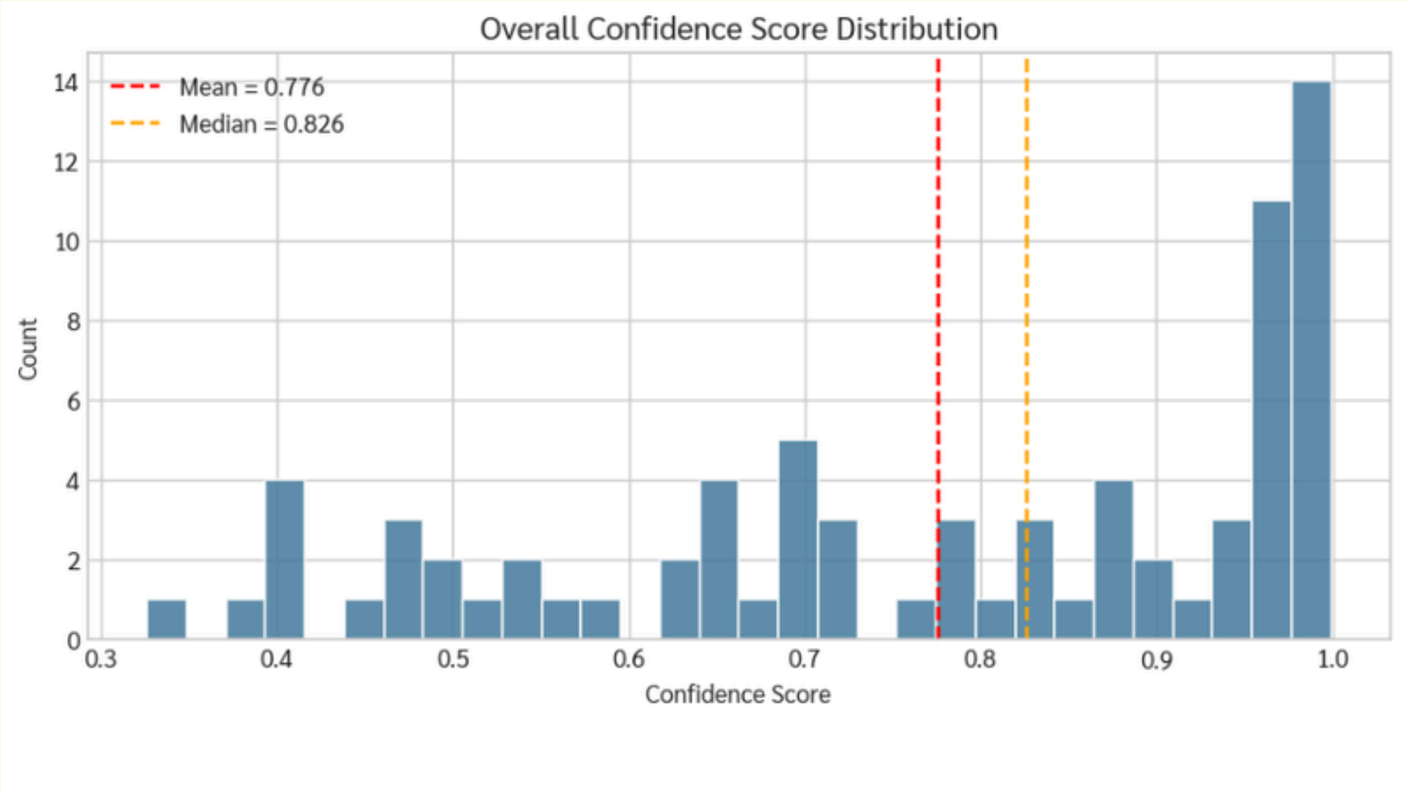
Confidence Score

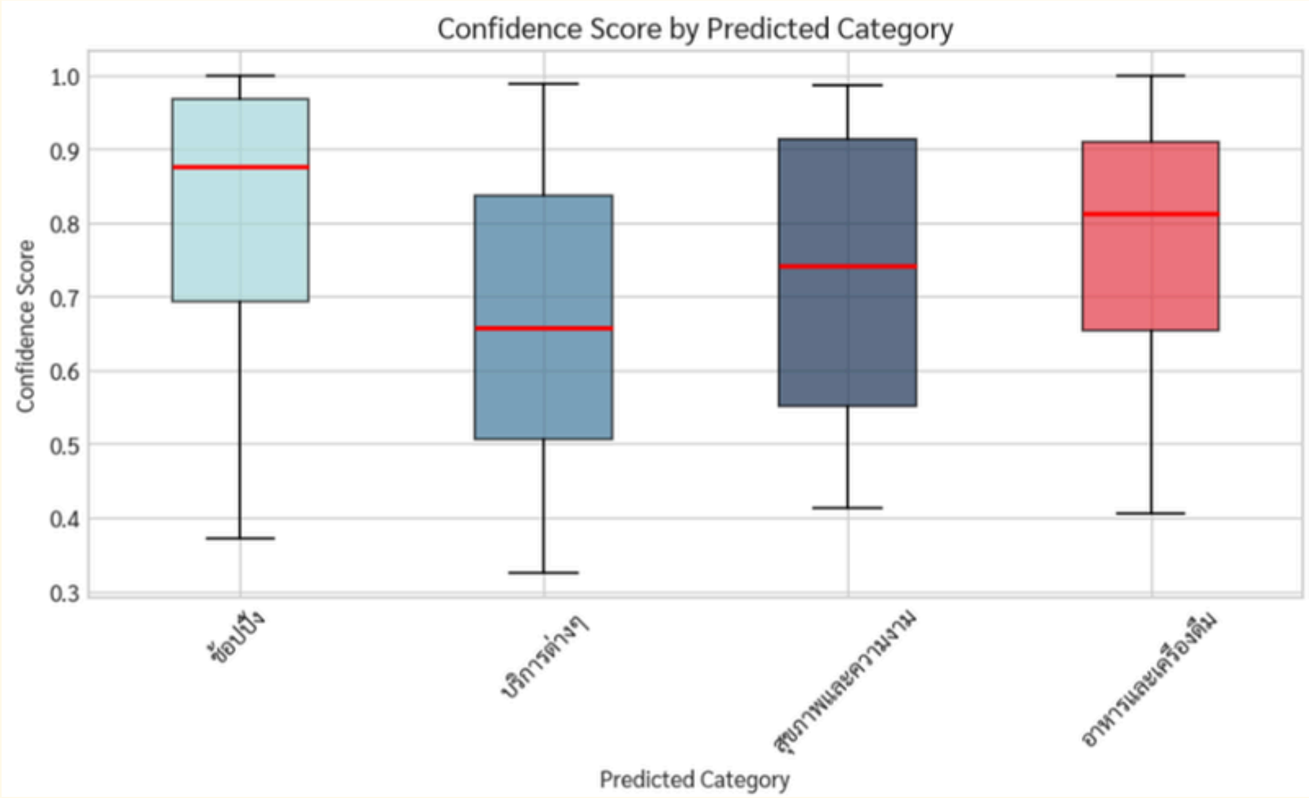


Problem **Solution** Insight Impact

Our Solution

Modeling: Hybrid: mCLIP + OCR model

Confidence Score



Problem **Solution** Insight Impact

Our Solution

Modeling: LLMs model

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LLMs model

Proprietary LLMs

- **Low** Data Privacy Control
- **Very-High** Performance
- **Expensive** Ongoing Cost

Open-Source LLMs

- **High** Setup Complexity
- **Complete** Data Control
- Very **Cost-Effective**

"Open-Source LLMs" is more suitable in this situation.

Problem | **Solution** | Insight | Impact

Our Solution

Modeling: Qwen2.5-VL model

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Qwen2.5-VL

What is Qwen2.5-VL?

- Agentic AI acts as a **Visual Agent**
- Preserving **tiny details**
- Specifically designed for **Finance & Commerce**

Why Qwen2.5-VL?

- higher **tolerance** for low-quality
- faster to **deploy and update**
- performs **visual reasoning**

How Qwen2.5-VL work?

- Uses a **specialized mathematic** to understand images
- **Read and reason** about a merchant's storefront
- **Compare** selected and predicted category

Problem **Solution** Insight Impact

Our Solution

Modeling: Qwen2.5-VL model

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Model Evaluation

Confusion Matrix

Actual \ Predicted	Consistent	Inconsistent
Consistent	49	24
Inconsistent	0	3

Accuracy: 0.6842

Weighted F2-score: 0.7053

Why F2-score?

Cost of Low Recall

- Inconsistent data may exist

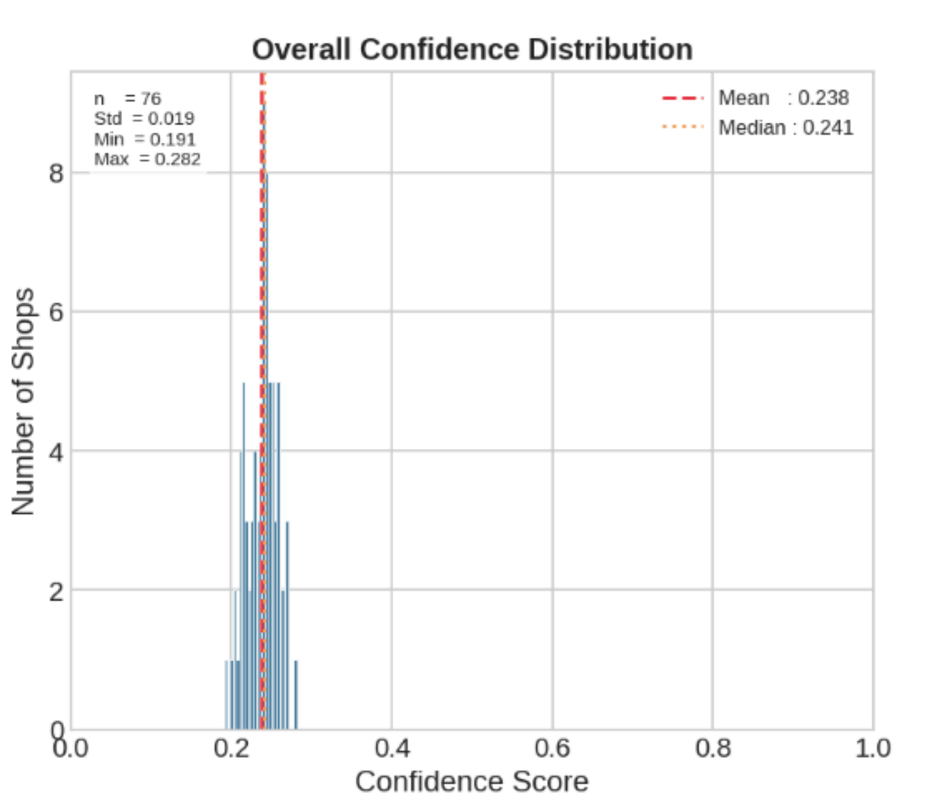
Cost of Low Precision

- Need Manual Review

Our Solution

Modeling: Qwen2.5-VL model

Confidence Score



Problem **Solution** Insight Impact

Our Solution

Modeling: Qwen2.5-VL model

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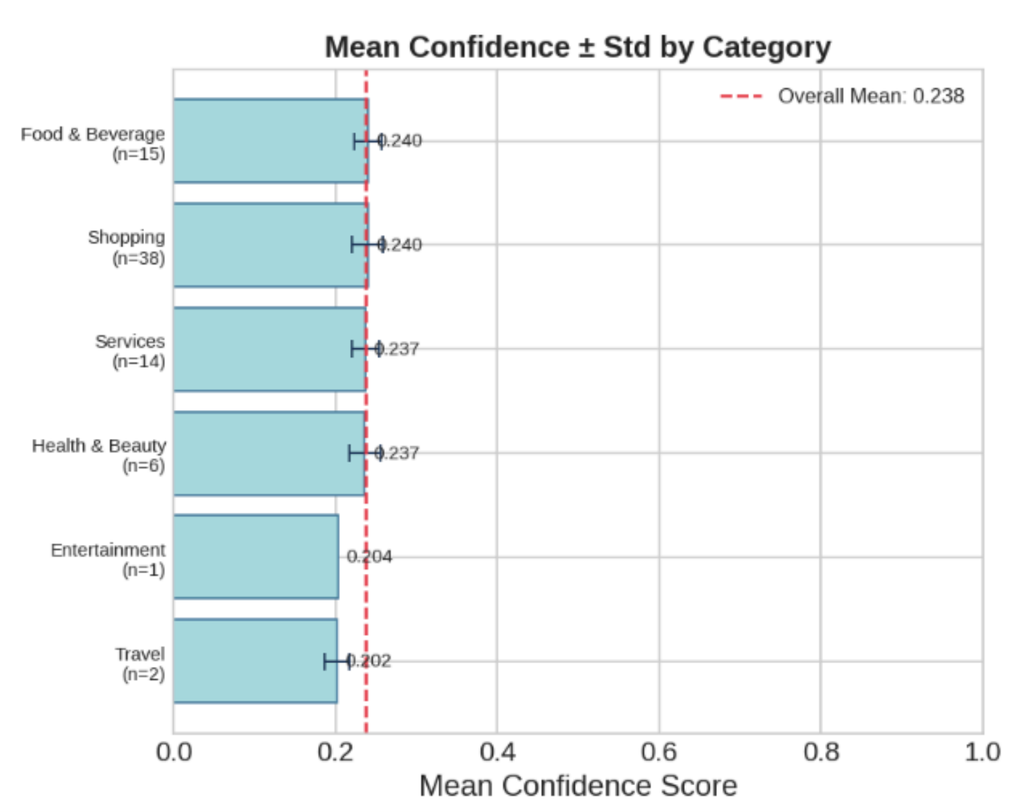
Confidence Score



Problem **Solution** Insight Impact

Model Selection

Criteria	mCLIP + OCR	Qwen2.5
Performance	4	5
Cost Efficiency	5	2
Latency	5	2
Explainability	3	5
Overall	**17**	**14**

"mCLIP + OCR model" is more suitable in this situation.

Demo mCLIP + OCR model

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Demo

What we've **got** from mCLIP + OCR model

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Error Analysis

Many of Error are from **Ambiguous Categories**

- Selected categories: **ช้อปปิ้ง**
- Predicted categories: **สุขภาพและความงาม**

It can be in both of Categories

Improvement

Create Specific & Strategic Category

- By creating a specific Pharmacy & Personal Care category, we **catch the in-between cases**

ช้อปปิ้ง → **เครื่องสำอางค์** / **ร้านขายยา**

Problem | Solution | **Insight** | Impact

What we've **got** from mCLIP + OCR model

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Performance Analysis		Insight
Shopping is strong but broad Highest accuracy but suffers lower precision because it absorbs errors from other categories.	→	**Redefine the Shopping Label** Clarify the 'Shopping' definition to reduce misclassifications.
Health & Beauty over-predicts Great recall but the worst precision on the board.	→	**Narrow the Health & Beauty definition** preventing general stores from being wrongly grouped here.
Services perform poorly The weakest category overall, with only 28.6% correct predictions and the lowest recall.	→	**Break Services into specific types** such as Laundry, Repair so merchants can easily identify their business.

Problem Solution **Insight** Impact

What **Insights** we have got?

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Business Insights

Anomaly Detection

Flags mismatches between declared and actual business types to prevent MDR arbitrage and policy abuse.

Granular Segmentation

Distinguishes micro-segments (e.g., Street Cart vs. Café) to enable precise credit limits and risk-based pricing.

Merchant Recommendation

Powers AI-driven, context-aware recommendations by utilizing verified merchant profiles to solve information overload, driving a 15-45% conversion uplift through precise **consumer matching**.

Problem Solution **Insight** Impact

What is the **business impact**?

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Business Impact

Operational Efficiency

instantly reduces manual review workload by 60-80%.

Credit Scoring & Lending

sharpens credit risk profiling to offer right-sized limits for micro-merchants.

Risk & Fraud Prevention

detects and blocks category gaming and fake shop registrations with high-precision visual audit.

Problem Solution Insight **Impact**

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THANK YOU

APPENDIX